CSC 309: Artificial Intelligence

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Week 4 : Rule Based System and Expert System

Week 4: Outline

- Rules for knowledge representation
- Rule Based (Production) System
  - Definition
  - Main components
  - Rule Execution Cycle
  - Conflict resolution
  - Applications
- Expert System

Rules for Knowledge Representation

- Rules are used as one way to represent knowledge.
- **Rules** are expressed in the following form:

If <conditions> then <conclusion>

If-pattern,
Premise
Or
Antecedent

Example:

If temperature is below 10

then weather is cold

Then-pattern,
Conclusion
or
Consequent

Rules for Knowledge Representation

- Rules are usually expressed in the form of IF . . . THEN . . . statements, such as:
- IF A THEN B
- A is called the **antecedent** and B is the **consequent** in this statement

  This can be considered to have a similar logical meaning as the following:
- $A \rightarrow B$

- In expressing rules, the consequent usually takes the form of an **action** or a **conclusion**.

Rules for Knowledge Representation

- Generally, a rule can have:
  - more than one antecedent, usually combined either by AND or by OR connective
  - more than one consequent, which usually suggests that there are multiple actions to be taken.
- The antecedent of a rule compares an **object** with a possible **value**, using an **operator**.
- For example, suitable antecedents in a rule might be:

  IF $x > 3$
  IF name is “Bob”
  IF weather is cold

- Here, the objects being considered are *x*, name, and weather; the operators are “>” and “is”, and the values are 3, “Bob,” and cold.

Examples of Rules

	Antecedents		Consequents
IF	Name is “ALi” AND Weather is cold	THEN	Tell Ali “Wear a Coat”
IF	saturday OR sunday	THEN	go to stadium
IF	NOT (saturday OR sunday)	THEN	go to work
IF	go to cinema	THEN	go outside
IF	go to work AND NOT at work	THEN	go outside
IF	NOT (can go outside)	THEN	stay home
IF	good weather	THEN	can go outside
IF	raining	THEN	have an umbrella
IF	raining AND have an umbrella	THEN	can go outside

Rule based Systems/Production Systems

- **Rule-based systems** or **production systems** are computer systems that use rules to provide recommendations or diagnoses, or to determine a course of action in a particular situation or to solve a particular problem.
- Instead of conventional algorithms, rule based systems use sets of IF–THEN rules (*production rules*).
- Unlike in algorithms, the order in which these rules should be used is not specified. It is decided by the program itself with respect to a problem state.
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Rule based Systems/Production Systems

- These systems are widely used in various domains, including expert systems, business process management, decision support systems, and more.
- Rule-Based Systems are designed to mimic human decision-making processes by encoding knowledge and expertise into a structured set of rules.
- In 1943, Post proved that any computable problem can be implemented in a production system.
- Cognitive scientists became interested in production systems because they seemed to represent better the way humans think and solve problems.

Components of Rule based Systems

- A rule-based system consists of a number of components:
- A database of rules (also called a **knowledge base**)
  - the knowledge base consists of a set of rules that represent the knowledge that the system has
- A database of facts (also called as working Memory)
  - The database of facts represents inputs to the system that are used to derive conclusions, or to cause actions
- An **interpreter**, or **inference engine**
  - The interpreter, or inference engine, is the part of the system that controls the process of deriving conclusions. It uses the rules and facts, and combines them together to draw conclusions.

Basic inference procedure in rule-based system

- The operation of a rule-based system follows a cyclic process:
  - **Matching:** The inference engine compares the conditions of each rule with the contents of the working memory to identify rules whose conditions are met.
  - **Selection:** Among the rules whose conditions are satisfied, the system selects the rules to be executed. This can involve prioritizing rules, resolving conflicts, and determining the appropriate sequence of rule execution.
  - **Execution:** The selected rules' associated actions are executed, leading to changes in the working memory or generating responses or outputs.
  - **Updating:** After rule execution, the working memory may be updated based on the changes brought about by the executed actions.
- The cycle then repeats as the system continues to process new information and make decisions.

Conflict Resolution Strategies

- In a situation where more than one conclusion can be deduced from a set of facts, there are a number of possible ways to decide which rule to fire (i.e., which conclusion to use or which course of action to take).
- For example, consider the following set of rules:

  IF it is cold
  THEN wear a coat
  IF it is cold
  THEN stay at home
  IF it is cold
  THEN turn on the heat

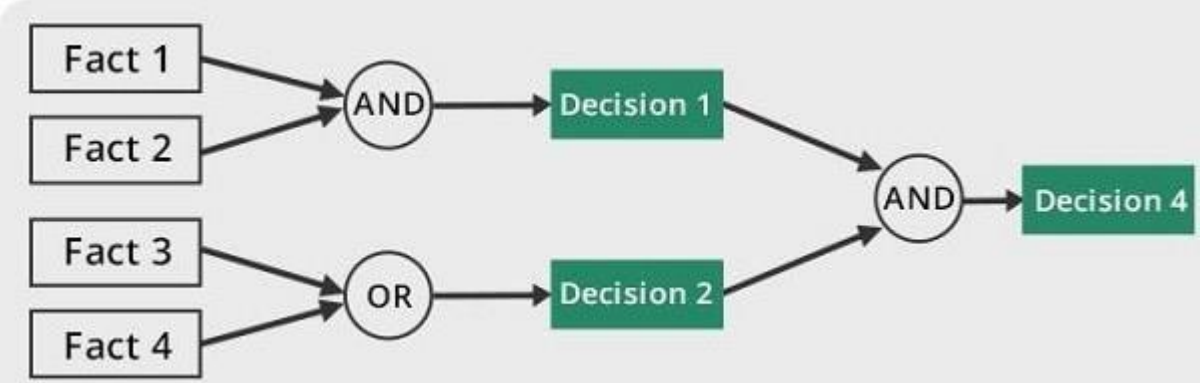
  If there is a single fact in the fact database, which is “it is cold,” then clearly there are three conclusions that can be derived

Conflict Resolution Strategies

- In some cases, it might be fine to follow all three conclusions, but in many cases the conclusions are incompatible (for example, when prescribing medicines to patients), and therefore the conflict need to be resolved using conflict resolution strategies
- **Conflict Resolution Strategies**
- **Refraction**
  - A rule can only be used once with the same set of facts in the working memory. This strategy prevents firing a single rule with the same facts over and over again
- **Explicit Priorities**
  - Use rules by their pre-defined order or priorities
- **Specificity**
  - Use the most specific rules
- **Recency**
  - Use rules that match the facts that were added most recently to the working memory, providing a kind of “focus of attention” strategy

General Methods of Inference in rule based system

- **Forward chaining**, also known as data-driven reasoning or bottom-up reasoning, involves starting with the available data or facts and applying rules to generate new conclusions or actions.
- the system starts from a set of facts, and a set of rules, and tries to find a way of using those rules and facts to deduce a conclusion or come up with a suitable course of action.
  - best for prognosis, monitoring, and control.



Forward Chaining Example

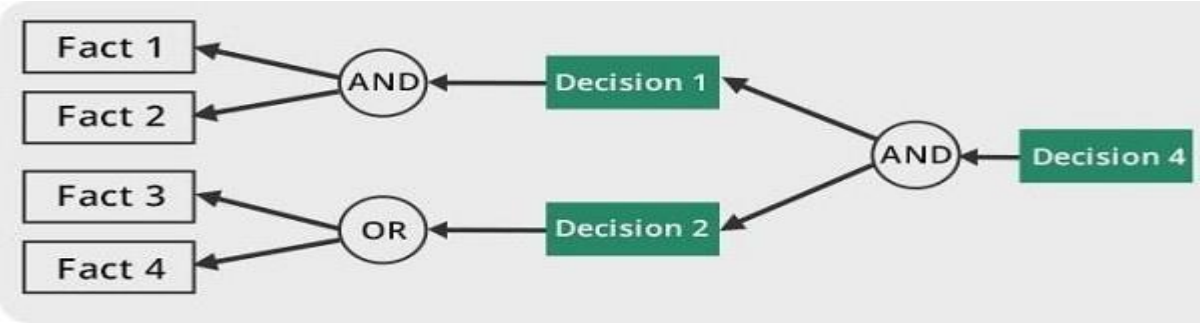
- **Example: Medical Diagnosis**
- Consider a medical diagnosis system that aims to identify a patient's illness based on their symptoms. The system has a set of rules that link specific symptoms to potential diseases. Here's how forward chaining works:
- **Initial Facts:** The system starts with the patient's symptoms as input data (e.g., fever, cough, headache).
- **Rule Application:** The rules are applied to the given symptoms. For instance:
  - If fever AND cough, then possibility of flu.
  - If headache AND runny nose, then possibility of sinusitis.
- **New Conclusions:** Based on the given symptoms and rule application, the system identifies potential diseases (e.g., flu, sinusitis).
- **Iteration:** If the system has more specific rules related to each potential disease, it can iteratively apply those rules to gather more evidence and refine the diagnosis.

Forward Chaining: Pros and Cons

- Advantages
  - It can be used to draw multiple conclusions.
  - It provides a good basis for arriving at conclusions.
  - It’s more flexible than backward chaining because it does not have a limitation on the data derived from it.
- Disadvantages
  - The process of forward chaining may be time-consuming. It may take a lot of time to eliminate and synchronize available data.
  - Unlike backward chaining, the explanation of facts or observations for this type of chaining is not very clear. The former uses a goal-driven method that arrives at conclusions efficiently.

Backward Chaining

- **Backward chaining,** also known as goal-driven reasoning or top-down reasoning, begins with a specific goal or conclusion and works backward through the rules to determine the necessary conditions or facts that must be satisfied to reach that goal.
- i.e. you starts from a conclusion/goal and move backward through the rules to identify the facts that led to that conclusion in the database of facts
  - best for diagnosis problems.



Backward Chaining Example

- **Example: Automated Diagnosis**
- Consider an automated diagnosis system for computer hardware issues. The goal is to identify the root cause of a malfunction.
- Rules:
  - IF no power THEN check power supply.
  - IF overheating THEN clean the fan and heat sink.
  - IF slow performance AND overheating THEN apply thermal paste.
- Goal: slow performance
  - Rule 3 is a possible solution for the goal.
  - Condition of rule 3 is not met (overheating), so we apply backward chaining to satisfy the condition.
  - Rule 2 addresses overheating.
  - Apply rule 2: overheating → clean fan and heat sink.
- The backward chaining process helps identify the necessary steps to address the goal.

Backward Chaining: Pros and Cons

- Advantages
  - The result is already known, which makes it easy to deduce inferences.
  - It’s a quicker method of reasoning than forward chaining because the endpoint is available.
  - In this type of chaining, correct solutions can be derived effectively if pre-determined rules are met by the inference engine.
- Disadvantages
  - The process of reasoning can only start if the endpoint is known.
  - It doesn’t deduce multiple solutions or answers.
  - It only derives data that is needed, which makes it less flexible than forward chaining.

Real-World Applications of Rule based System

- **Expert Systems:** Rule-based systems are extensively used to build expert systems that replicate the decision-making abilities of human experts in various fields, such as medicine, finance, and engineering.
- **Business Rules Management:** Organizations use rule-based systems to automate and manage business processes, ensuring compliance with regulations and optimizing operations.
- **Diagnosis and Troubleshooting:** Rule-based systems can assist in diagnosing problems and suggesting solutions in technical support scenarios.
- **Decision Support:** Rule-based systems aid in making informed decisions by providing recommendations based on predefined rules.

Expert Systems

What is Expert System (ES) ?

- ES perform tasks for specific domains that require human expertise
  - Examples: Medical diagnosis, fault diagnosis, status monitoring, etc.
- ES solve problems using domain-specific knowledge
  - Domain knowledge is acquired by interviewing human experts
- ES were the first successful applications of AI to real–world problems solving in medicine, chemistry, finance and even in space (Space Shuttle, robots on other planets).

Some Definitions of “ES”

- A computer system that emulates the decision-making ability of a human expert in a restricted domain [Giarratano & Riley 1998]
- “An intelligent computer program that uses knowledge and inference procedures to solve problems that are difficult enough to require significant human expertise for their solutions.” [Giarratano & Riley 1998]
- Computer programs that try to replicate knowledge and skills of human experts in some area, and then solve problems in this area (the way human experts would).

Why ES?

- Reproduce Expertise: make available
- Record Expertise: experts die
- Combine expertise of many
  - Didn’t work
- Teach expertise
  - Didn’t work
- Expand application area of computers
  - What tasks can computers do?

Some Famous Early ESs

- Dendral (Stanford, Ed Feigenbaum, 1965) - organic chemical analysis
- Macsyma (MIT, 1965) - symbolic math problems
- Mycin (Stanford, 1972) - diagnose blood diseases
- Prospector (SRI, 1972) - mineral exploration
- Caduceus (1975) - internal medicine
- Xcon and R1 (1980, 1982) - computer system configuration (for DEC)
- Harpy - document retrieval
- Hearsay - Speech understanding
- Max (Nynex, 1992) - telephone network troubleshooting bacterial infection diagnosis

Main Components of an ES

- knowledge base
  - contains essential information about the problem domain
  - often represented as facts and rules
- inference engine
  - mechanism to derive new knowledge from the knowledge base and the information provided by the user
  - often based on the use of rules
- user interface
  - interaction with end users
  - development and maintenance of the knowledge base

Main Components of an ES

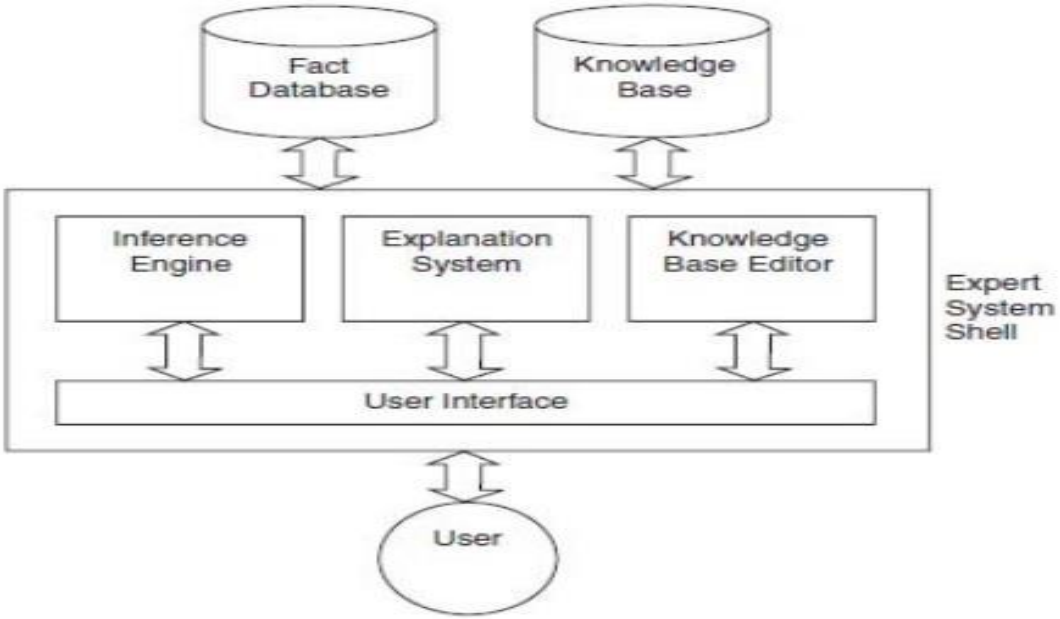


Fig Expert System Architecture

Main Components of an ES

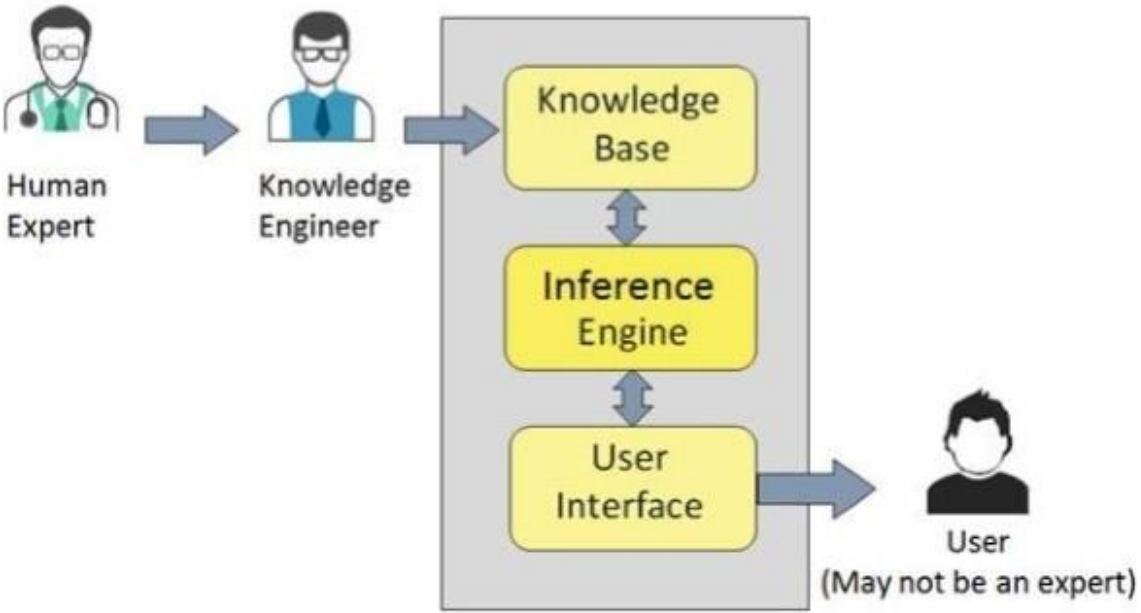
- Fact database also know as working memory
  - The fact database represents input to the system that are used to drive new conclusions
  - Hold the current state of the system including data and information needed for making decisions
- Explanation system
  - The explanation system provides information to the user about how the inference engine arrived at its conclusions.
- The knowledge-base editor
  - Allows the user to edit the information that is contained in the knowledge base.
  - It is used by knowledge engineer or the expert to provide and update the knowledge that is contained within the system

Main Components of an ES

- **The Expert System Shell**
  - The parts of the expert system that do not contain domain specific information are contained within the expert system shell
  - It is a general framework that can be used to build a number of different expert systems, depending on which KB is added to the shell.

- Typical example of such a shell is CLIPS (C Language Integrated Production System),
- Other examples in common use include OPS5, ART, JESS, and Eclipse.

People involved in an ES



People involved in an ES

- The design, development, and use of expert systems involves a number of people.
- **End-User**
  - The **end-user** of the system is the person who has the need for the system.
  - In the case of a medical diagnosis system, this may be a doctor, or it may be an individual who has a complaint that they wish to diagnose.
- **Knowledge Engineer**
  - The **knowledge engineer** is the person who designs the rules for the system, based on either observing the expert at work or by asking the expert questions about how he or she works.
- **Domain Expert**
  - The **domain expert** is very important to the design of an expert system.
  - In the case of a medical diagnosis system, the expert needs to be able to explain to the knowledge engineer how he or she goes about diagnosing illnesses

When (Not) to use an ES?

- Expert systems are not suitable for all types of domains and tasks
  - conventional algorithms are known and efficient
  - the main challenge is computation, not knowledge
  - knowledge cannot be captured easily
  - users may be reluctant to apply an expert system to a critical task

Lecture 5 Review Questions

- Expert System and Rule Based Systems
  - Explain why expert systems are so called.
  - Explain the difference between forward chaining and backward chaining. Explain the advantages and disadvantages of each method.
  - Explain the various methods of conflict resolution that can be used in rule-based expert systems. For each of these, give an example of a scenario where using it would *not* give the correct result.
  - What is the purpose of Meta rules? Would an expert system have any advantages if it knew the difference between Meta rules and normal rules?
  - Describe the architecture of an expert system, and describe the roles of the various people involved in it

Credits/References

- Artificial Intelligence Illuminated
  - **By Ben Coppin**
- Artificial Intelligence: A Modern Approach
  - **By Stuart Russell and Peter Norvig**
- Internet
- Fppt.com